

EXP 1 : First Palindromic String in an Array

Aim :

To write a Python program to find the **first palindromic string** in an array of strings and analyze its time complexity.

Algorithm :

1. Start.
2. Read an array of strings.
3. Traverse each string in the array from left to right.
4. Check whether the current string is equal to its reverse.
5. If it is a palindrome, display it and stop the search.
6. If no palindromic string is found, display an empty result.
7. Stop.

Source Code :

```
def is_palindrome(s):  
  
    return s == s[::-1]  
  
def first_palindrome(words):  
  
    for word in words:  
  
        if is_palindrome(word):  
  
            return word  
  
    return ""  
  
words = ["abc", "car", "ada", "racecar", "cool"]  
  
result = first_palindrome(words)  
  
print("Array:", words)  
  
print("First Palindromic String:", result)
```

Output :

```
Output
Array: ['abc', 'car', 'ada', 'racecar', 'cool']
First Palindromic String: ada
```

Result :

Thus, the Python program was successfully executed and the first palindromic string in the given array, “**ada**”, was identified.

Experiment 2: Common Elements Between Two Arrays

Aim

To write a Python program to find the **common elements between two arrays** and analyze its time complexity.

Algorithm

1. Start.
2. Read two arrays.
3. Convert the second array into a set.
4. Traverse each element of the first array.
5. Check whether the element exists in the set.
6. If it exists, add it to the result.
7. Display the common elements.
8. Stop.

Source Code

```
def common_elements(arr1, arr2):  
    set2 = set(arr2)  
    result = []  
  
    for element in arr1:  
        if element in set2 and element not in result:  
            result.append(element)  
  
    return result  
  
arr1 = [1, 2, 3, 4, 5]  
arr2 = [3, 4, 5, 6, 7]  
  
result = common_elements(arr1, arr2)  
  
print("Array 1:", arr1)  
print("Array 2:", arr2)  
print("Common Elements:", result)
```

OUTPUT :

```
Array 1: [1, 2, 3, 4, 5]  
Array 2: [3, 4, 5, 6, 7]  
Common Elements: [3, 4, 5]
```

RESULT :

Thus, the Python program was successfully executed and the common elements between the two given arrays, **3, 4, and 5**, were identified.